Number Theory Tutorial

Orders and Primitive Roots

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Problems

1. **Computing Orders.** Find the order of each of the following elements:
   (a) 2 modulo 13;
   (b) 3 modulo 17;
   (c) 5 modulo 23.
2. **Orders of Powers.** Find the order of 7 modulo 40. Deduce the order of 7^3 modulo 40.
3. **Elements of a Given Order.** Determine all elements of order 4 in $(\mathbb{Z}/17\mathbb{Z})^\times$.
4. **Order in a Cyclic Group.** Find all integers a with $1 \leq a < 19$ such that the order of a modulo 19 is 6.
5. **Testing for Primitive Roots.** Determine whether 2 is a primitive root modulo 17. Justify your answer by computing suitable powers of 2 modulo 17.
6. **Finding Primitive Roots.** Find all primitive roots modulo 11.
7. **Testing a Candidate.** Determine whether 3 is a primitive root modulo 19. If it is, find all primitive roots modulo 19.
8. **Smallest Primitive Root.** Find the smallest positive primitive root modulo 23. Prove that your answer is correct.
9. **All Primitive Roots.** Determine all integers n with $1 \leq n < 29$ such that n is a primitive root modulo 29.